

STATEMENT OF PURPOSE

Srashti Katiyar

B.Tech. (Computer Science – Artificial Intelligence) | Banasthali Vidyapith | CGPA: 9.19/10

Academic Background and Intellectual Development

I am currently in my third year pursuing my B.Tech. in Computer Science and Artificial Intelligence. My academic interests lie in the core computer science fundamentals. Throughout my academic career, subjects like Data Structures, Design and Analysis of Algorithms, Database Management Systems, Operating Systems, and Software Engineering have helped me develop a structured and analytical way of approaching problems.

I find the study of the application and practicality of abstract computer science theories to efficient systems particularly interesting. I have also had the benefit of studying the various paradigms and complexities of algorithms. I have also had the benefit of studying subjects like operating systems and database management systems.

Technical Exposure and Practical Learning

Apart from the theoretical knowledge that I have gained from my curriculum, I have also been actively engaged in practical learning through the projects that I have developed so far. Some of the activities that I have been actively engaged in include coding practice to improve my logical reasoning skills.

Technical Skills and Applied Learning

In addition to academic learning, I focus on applying learning through actual implementation and execution of concepts and theories learned. For example, I have been working on improving my skills in algorithm-based problem-solving and have been applying it to enhance my critical thinking skills and improve efficiency in problem-solving algorithms.

One of the most significant learning experiences I have had was working on my ongoing project titled “LifeLink – Every Second Counts.” It is a web-based platform designed to ease and streamline medical fund assistance by providing a workflow between patients, hospitals, and NGOs. In working on this project, I was tasked with developing backend logic, authentication logic, structuring workflow, and integrating payment API.

In working on this project, I have learned a great deal about how real-world problems and issues have significant implications on how one applies theoretical learning and understanding of system design and software engineering principles and how they must adapt and respond to real-world issues such as latency, consistency, and usability.

Research Motivation and Areas of Interest

My academic and research-based project experiences have gradually influenced my research-based problem-solving interests in the computer science domain. My research interests include system optimization, scalable backend systems, distributed computing fundamentals, and efficient data processing techniques.

I am also curious to know and learn about the process of evaluating large systems for performance and efficiency and the role research plays in enhancing the efficiency and performance of systems.

Why IIT Madras?

IIT Madras is an intellectually rigorous and research-based academic institution that meets my academic and research-based goals and interests. IIT Madras provides the best academic platform for academic growth and research-based problem-solving.

Through this academic experience at IIT Madras, I am hoping to get research-based exposure and to learn the formalities and methodologies of experimentation and evaluation. I also believe this academic experience would provide me the platform to make significant contributions to research-based projects.

I am highly motivated and adaptable to new situations and academic environments. I believe this academic experience at IIT Madras would greatly enhance my knowledge and understanding of advanced computer systems and would play an important role in my future research and academic contributions to the computer science and research-based problem-solving fields.